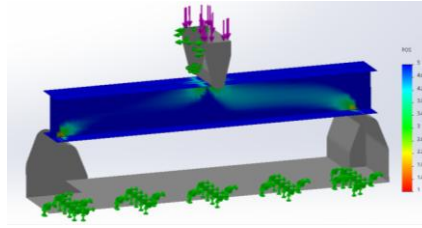
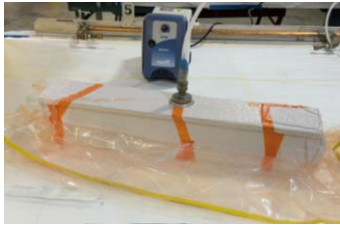


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CARBON FIBER I-BEAM - SAMPE BEAM COMPETITION PRACTICE



What?

- Advanced composite manufacturing
- Determine optimal layup schedule for 7200 lbf design load
- Iterate design to improve load at break

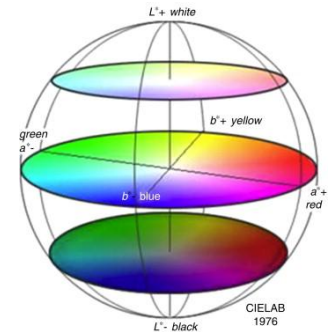
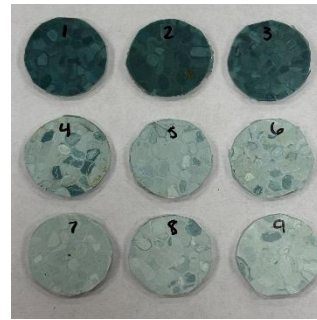
How?

- UD + plain weave CF **prepreg**
- Used **Excel** for force calculations
- **Autoclave** for ramp temp cure
- Compared **FEA** vs **UTM failure load**

Results

- Increased MTS load frame **failure load** by **554%** (1471 to **9622 lbf**)
- Exceeded 7200 lbf design load
- **Specific strength** of 17.81 lbf/g

HDPE COLOR COMPOUNDING - OCEAN PLASTICS RECOVERY (OPR)



What?

- Blend mixed-color ocean-captured plastic into uniform shade feedstock
- Minimize % pigment to achieve shade uniformity
- Sustain **thermomechanical properties**

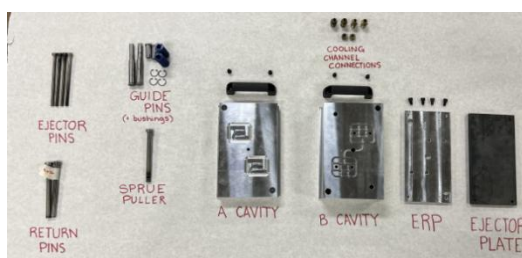
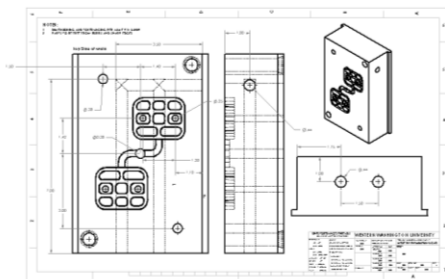
How?

- Blend HDPE with TiO_2 pigment in **twin screw extruder**
- Vary TiO_2 % and screw speed (rpm) with 3^2 designed experiment
- **Colorimetry**, **rheology**, and **DMA**

Results

- Optimal conditions assessed to be 2% TiO_2 , 300rpm
- No convincing effect from rpm or TiO_2 % on batch variability, viscosity or modulus

INJECTION MOLD FABRICATION - OCEAN PLASTICS RECOVERY (OPR)



What?

- Continuation of compounding project
- **Injection mold fab** for production of pocket-sized watercolor palettes

How?

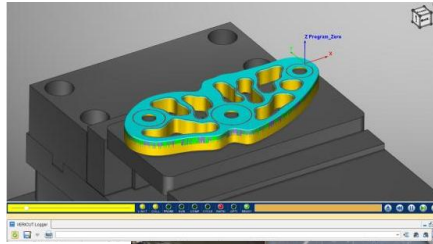
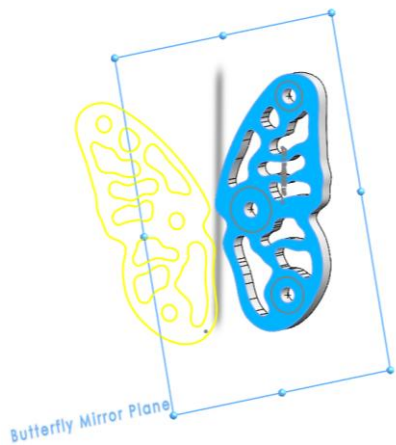
- Used **design for manufacturing** skills to ensure manufacturability with CNC tools
- **Prototyped** and iterated with **SolidWorks** and **3D printing** to align on design with customer
- Designed **MUD frame cavity-core injection mold** with **SolidWorks 3D CAD** and 1:1 sized **2D engineering drawings**
- Machined cavity, core, water lines, taps with **Haas VF2 mill**
- Installed mold, successfully produced 25 parts

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BIKE REAR SUSPENSION LINK (MONARCH WING)



What?

- Design a creative bike link for rear suspension with constraints of bore locations and diameters
- Program, set up, and operate mill independently, demonstrating growth in machining knowledge

How?

- Design in **SolidWorks**, program in **HSMWorks** with HSM and HEM techniques
- Post process in **VERICUT**
- Set up and machine Op1 and Op2 in **Haas VF2 mill**

Results

- Design showcased creativity and understanding of 2.5 axis machining
- End-to-end project responsibility cemented understanding of how design, tolerances, and specifications impact work on the machining shop floor

CARBON FIBER AND KEVLAR COMPOSITE BODICES



Female tooling
(interior surface)



What?

- Deepen hands-on composite manufacturing skills
- Create a custom composite product with **female tooling**
- Inspired by Mugler's Maschinenmensch, Issey Miyake's "Plastic Body" and Ultranova's breastplates

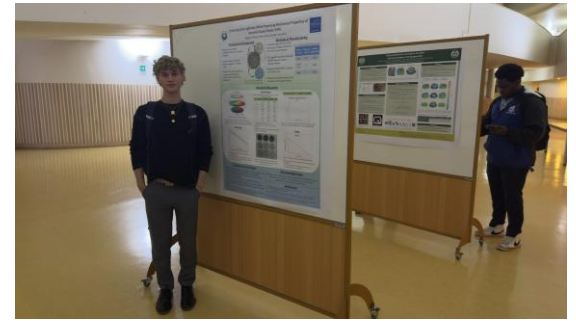
How?

- Source mannequin for tooling, critically, made from a polyolefin for ease of release from tooling surface
- Build custom fixture
- Lay up 8 plies **satín weave** carbon fiber for ease of drape over complex curves
- Apply epoxy, cure under vacuum

Results

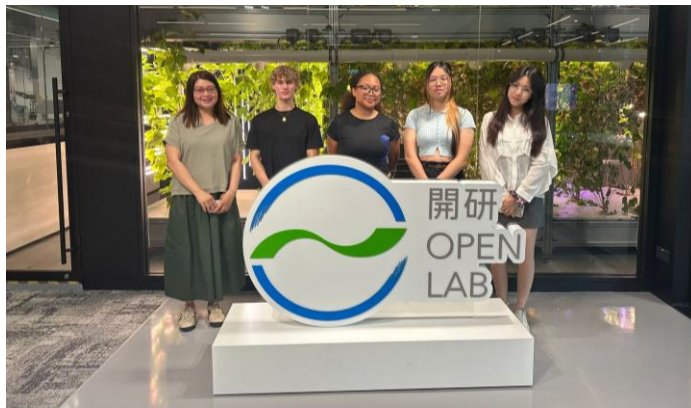
- Manufactured two form-fitting composite bodices
- Improved surface wet-out on second iteration (Kevlar bodice) by increasing vacuum
- Demonstrated ability to move concept to finished product
- Bridged engineering and arts

BIOINSPIRED DESIGN IRES-BioFLOW AWARD (SANTIAGO, CHILE)



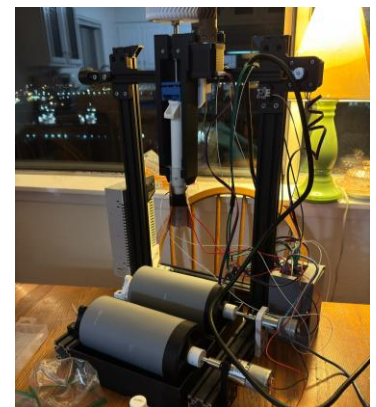
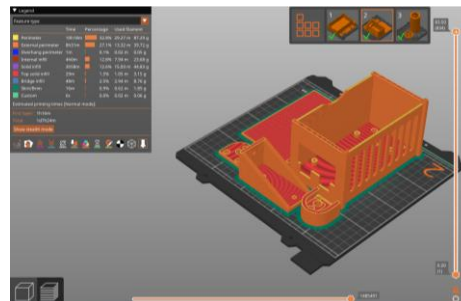
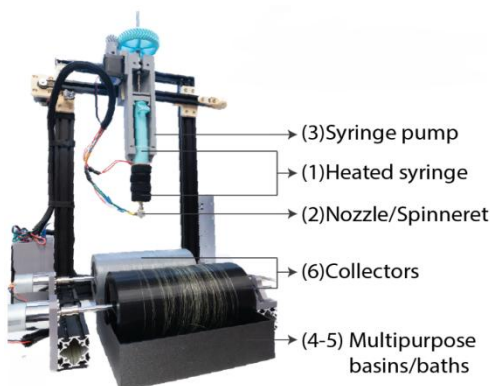
Awarded fully-funded opportunity to present research at [culminating Bio-FLOW IRES event](#) at Universidad de Adolfo Ibañez in **May 2025**.

OPEN LAB GLOBAL CHALLENGE (HONG KONG, SAR, CHINA)



Received first-place award by Hong Kong Research Institute of Textiles and Apparel in [inaugural Open Lab Global Challenge](#), for technical concept development **of materials engineering solution** through **design for circularity** lens (with **life cycle analysis**) to tackle **textile waste** generated by premature deterioration of socks. Honored at award ceremony and Innovation & Technology Symposium in Hong Kong in **July 2025**.

WWU UROP RESEARCH GRANT (BELLINGHAM, USA)



Received funding award for grant application in **March 2025** to build an open-source desktop-sized biofibers spinning machine for the advancement of the WWU Polymer Materials Engineering program and to increase hands-on student engagement with bioinspired design, production processes, and textiles and fibers.



Twists & Torsions: SAMPE Carbon Fiber I-Beam

Ethan Granat, Rachel Heinke, Lila Rockwood | 2 December 2024
Polymer Materials Engineering, Western Washington University



Purpose

In this project, we set out to design, manufacture, and break (under 3-point bending load) a lightweight carbon fiber I-beam. The 2024 rules for Category A of the SAMPE student bridge competition were followed to meet a minimum load of 1500lbf, and a design load goal of 7200lbf [1]. The purpose was to gain experience and confidence in self-directed project management of advanced composite part development.

Materials

Toray plain weave CF-epoxy prepreg
T830H-6K fiber, 3900 series epoxy [2]
Toray unidirectional CF-epoxy prepreg
T800S-24K fiber, 3900 series epoxy [2]

Aluminum tooling

- Two flat flanges (top and bottom)
- Two C-channels

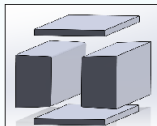


Figure 1: I Beam tooling exploded view.

Consumables

Vacuum bagging, vacuum tape, bleeder, and peel ply.

Autoclave Cure Cycle

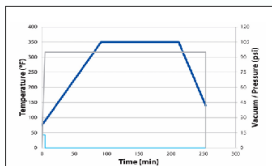


Figure 2: Manufacturer's recommended autoclave cure cycle [2].

Design

- Plain weave only
- Four plies per flange
- Four plies per C-channel
- Eight total plies thick in web
- Eight total plies thick in top/bottom faces.
- Plain weave created internal symmetry.

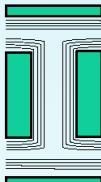


Figure 3: Beam 1 layup (4 plies per C-channel, and 4 plies per flange).

Beam I

Results

Load Testing Results

- Part weight: 0.52kg
 - Load at break: 1471 lbf
 - 2.83 strength-to-weight ratio (lbf/grams)
- Beam 1 showed bending-twisting failure, didn't meet minimum load requirement.

Part Quality Results

- Void in middle of I cross-section (Figure 4).
- Deep wrinkles on top/bottom flanges.



Figure 4: Void observed in cross-section (noodle) of first beam.

Manufacturing Steps



Figure 8: Kit cutting.



Figure 9: Layup.

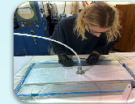


Figure 10: Debulking.

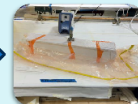


Figure 11: Assembled and bagged.

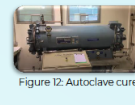


Figure 12: Autoclave cure.

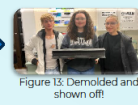


Figure 13: Demolded and shown off!

Beam II

Iteration

- Many more plies added:
 - 16 plies per C-channel.
 - 8 plies per flange [X].
- Addition of 45° plies in C-channels.
- Unidirectional plies added in C-channels to increase strength in direction of loading.
- Noodle made from unidirectional (rather than plain weave) to increase noodle compaction.
- Width trimmed to 3" wide.
- Puck moved from center to edge of beam (later trimmed off) to decrease wrinkling.



Figure 5: Beam 2 layup, with a total web thickness of 32 plies and total flange thickness of 24 plies.

FEA of Improved Layup Design

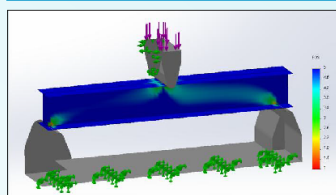


Figure 6: FEA factor of safety plot. Beam 2

FEA model of second beam exceeded 7200lbf design load with min FOS of 1.39.

Results

Load Testing Results

- Part weight: 0.54kg
- Load at break: 9622 lbf
- 17.81 strength-to-weight ratio (lbf/grams)

Part Quality Results

- Excess resin flow-difficult to debag
- Much smoother flange surface finish

Beam 2 failed between web and bottom flange. Failure load was 6.5x greater failure load than Beam 1, with only 3.85% increase in weight, due to addition of unidirectional plies in web.



Figure 7: Beam 2 after trimming.

Future work

Future work in this project should involve replacing more woven web plies with unidirectional plies and performing an FEA-based designed experiment on layup sequence—both for the purposes of optimizing strength-to-weight ratio.

Acknowledgments

We'd like to thank Nikki Larson, Russell Choi-Pearce, and Leia Kaminsky for their support and guidance throughout this project.

References

- [1] "Student Bridge Contest," SAMPE Conference, 2024. Available: <https://www.sampeamerica.org/student-bridge-contest>
- [2] Toray Composite Materials America, "3900 Prepreg System Data Sheet," Jul. 2020. Available: <https://www.toraycma.com/wp-content/uploads/3900-Prepreg-System.pdf>

Introduction & Background

- Marine-sourced HDPE (opHDPE) is found in a myriad of colors
- Lighter shades of recycled plastic are higher in market value¹
- Pigment TiO_2 was explored as a method of brightening opHDPE while maintaining properties²

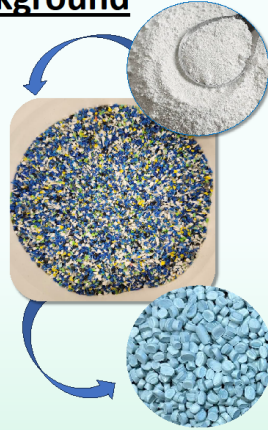


Figure 1. Depiction of opHDPE color change through compounding of TiO_2 .

Methods & Manufacturing

- Compounding of the material took place on the Lab Tech 16 mm twin screw extruder
- TiO_2 pigment was hand fed and opHDPE was fed using a vibratory feeder
- Designed experiment: 2 factors and 3 levels (3^2)
- Responses measured were color, viscosity, and storage modulus

Table 1. Designed experiment real variables.

Factor Level	TiO_2 wt % (C)	Screw Speed rpm (N)
Low	0.5	200
Center Point	2	300
High	5	400

Results & Discussion

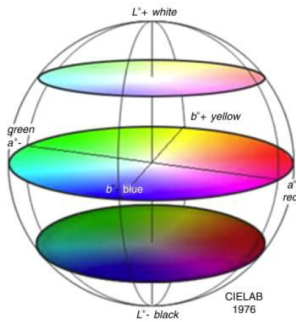


Figure 2. CIE $L^*a^*b^*$ colorimeter scale [3].

The CIE $L^*a^*b^*$ color space is a 3D color space which assigns absolute numerical values to color in terms of light to dark (L^*), green to red (a^*), and blue to yellow (b^*) [3].

Colorimetry

Table 2. Standard deviations (σ) of L^* , a^* , and b^* color space values.

DOE Run	σ of L^*	σ of a^*	σ of b^*
1	4.31	0.82	0.21
2	3.10	0.95	0.53
3	3.53	0.83	1.73
4	3.30	0.20	1.04
5	9.68	0.60	0.84
6	4.51	1.02	1.12
7	1.71	0.16	0.31
8	2.24	0.44	1.07
9	0.94	0.10	0.39

- The greatest deviation in each run corresponded to the lightness (L^*) of the samples.
- Red-green variation (a^*) was linearly correlated with TiO_2 concentration.

$$\text{Std. Dev. of } a^* = -0.318 C + 0.5697$$

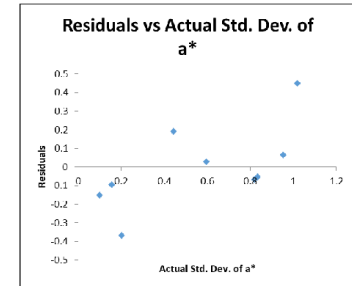


Figure 3. Non-random residuals of linear regression model relating standard deviation of a^* to TiO_2 concentration.

Rheology

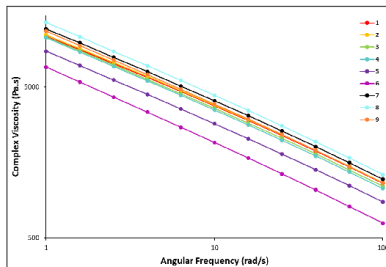


Figure 4. Rheology viscosity (Pa-s) curves.

- Rheology Experiment: Isothermal Shear Sweep
 - Temperature: 215 °C
- Rheology demonstrated that concentration of TiO_2 pigment had little impact on viscosity as no consistent trend was observed between runs.

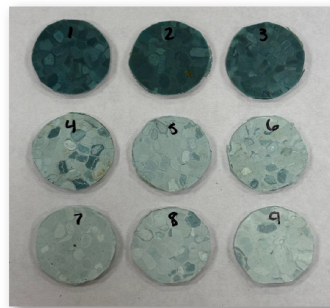


Figure 5. Rheology samples exhibiting color change through batches (top left: DOE run 1; bottom right: DOE run 9).

DMA

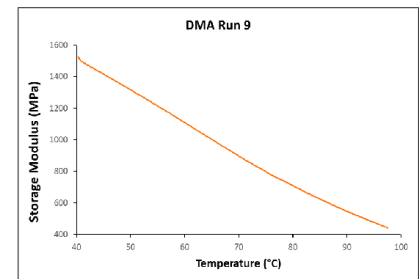


Figure 6. Storage modulus E' (MPa) from 40-100°C.

- Storage modulus was analyzed at 70 °C (temperature midpoint).
- Results revealed that storage modulus did not correlate with TiO_2 concentration.

Conclusion & Future Work

- The standard deviation of a^* (red to green color) was found to have a significant relationship ($p < 0.05$) with TiO_2 concentration.
- Viscosity was found to significantly correlate ($p < 0.05$) with the square of TiO_2 concentration.
- Residual plots are unconvincing due to lack of replicate tests.
- Future work should involve repeating the compounding process while using the continuous gravimetric feeder (as opposed to hand feeding).
- DMA and rheology should be performed in triplicate for each DoE run to increase accuracy of linear regression models.

Acknowledgements

Thank you to Dr. John Misasi, Leia Kaminsky, and Russell Choi-Pierce for assistance during the compounding process, and to Kyle Mikkelsen for characterization assistance.

References

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- [2] A. Turner and M. Filella, "The role of titanium dioxide on the behaviour and fate of plastics in the aquatic environment," *Science of The Total Environment*, vol. 869, p. 161727, Apr. 2023.
- [3] "Method of Measuring and Specifying Colour Rendering Properties of Light Sources," *Journal of the Illuminating Engineering Institute of Japan*, vol. 49, no. 10, pp. 585-604, 1965.

Background & Motivation

Problems & Needs

America sends 11.3 million tons of textile waste to landfill annually. 85% of all textiles go to landfill, whereas only 15% ever recirculate in the economy [1]. Developing novel methods and materials to design textiles with viable end-of-life pathways *prior to the point of manufacturing* is critical to reducing textile and fashion waste.



Figure 1. Textile waste landfill in the Chilean Atacama Desert, receiving 56,000 tons of clothing waste per year [2, 3].

Solutions

The desktop biofibers spinning machine is one such solution to this need. By building a small-scale, low-cost, and open-source machine to spin biofibers, the barrier to experimentation and implementation of novel and sustainable textile/fiber design strategies is significantly reduced, allowing researchers at any level to tackle textile waste issues.

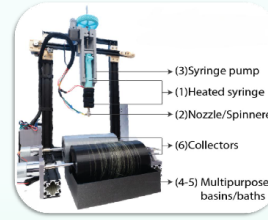


Figure 2. Original assembled desktop biofibers spinning machine built by CU Boulder ATLAS Institute researchers [4].

Outcomes

The machine is akin to a 3D printer, but instead of printing solid parts, it spins flexible, continuous, biobased fibers which can be woven, knitted, or even incorporated into smart textiles. Once no longer useful, the biofibers can be selectively dissolved, composted, or otherwise recirculated through intentional design for circularity.

Project Overview

Machine Assembly

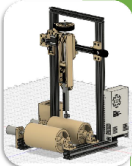


Figure 3. CU ATLAS spinning machine CAD model in Fusion [4].

- Source components
- Print 3D printed parts
 - Assemble frame
- Assemble subsystems
- Wire electronic circuits
- Connect all subsystems

Fiber Formulation Development



Figure 4. Squid ring teeth (SRT) protein solution [5].

- Perform literature review to identify candidate materials (e.g.,
- Choose polymeric biofiber reagents for selective dissolution
- Synthesize formulation

Fabrication & Iteration



Figure 5. Lustrous biofibers spun by CU ATLAS team [4].

- Spin fiber formulation
- Weave or knit fibers into textile
- Test structure for ability to selectively dissolve
- Iterate formulation through designed experiment methodology

Progress

Over the course of four weeks, all necessary pre-fabricated components have been identified, ordered, and received. The majority of the custom 3D-printed components have been successfully sliced and printed. The machine frame has been assembled.

Challenges

Numerous components of the biofibers spinning machine are custom-designed 3D-printed parts. Slicing issues, print failures, and unexpected costs beyond the budget limit have resulted in a 4-week delay beyond the anticipated timeline.

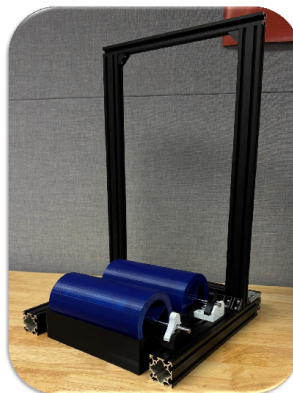


Figure 6. Assembled frame, spindle collectors, and collector trays.

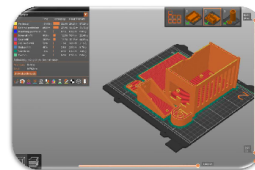


Figure 7. Sliced part models in PrusaSlicer.



Figure 8. 3D-printed tensioner screw cap: failed print (left) and successful print (right).

Future Work

- | Short-Term | Long-Term |
|---|---------------------------------------|
| • Identify project successors | • Assemble remaining subsystems |
| • Perform root-cause analysis of repeated print failures on | • Fully assemble machine |
| • Print remaining parts | • Experiment formulations of interest |
| • Assembly LVE syringe | • Permanently house in WWU Makerspace |
| • Compile legacy document | • Create user guide |

Acknowledgements

Thank you to Dr. John Misasi and Dr. Sura Al-Qudah for your support in nominating this project for funding. Thank you to Leia Kaminsky for supply sourcing, to the STC for 3D printing support, and thank you to Dr. Michael Rivera and the rest of the CU Boulder ATLAS team for designing and publishing the original open-source work. Finally, thank you to the WWU Undergraduate Research and Creative Opportunities Grant committee for funding this project.

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- [2] "SkyFi's Satellite Image Confirms Massive Clothes Pile in Chile's Atacama Desert | SkyFi," *Skyfi.com*, 2022. Available: <https://skyfi.com/en/blog/skyfis-confirms-massive-clothes-pile-in-chile>
- [3] C. Sánchez, "Atacama Desert's Solastalgia: Color and Water for Dumping," *Biosemiotics*, Jan. 2024, doi: <https://doi.org/10.1007/s12304-023-09551-w>
- [4] L. Vasquez, Mirela Alistar, L. Devendorf, and M. L. Rivera, "Desktop Biofibers Spinning: An Open-Source Machine for Exploring Biobased Fibers and Their Application Towards Sustainable Smart Textile Design," pp. 1–18, May 2024, doi: <https://doi.org/10.1145/3613904.3642387>
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Optimizing Cure Conditions for Frontal Polymerization Composites

Ethan Granat, Anders Johnson, Dr. Mark Peyron

June 2025



Background

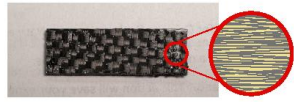


Figure 1. Carbon fiber and RapiCure R2 resin composite.

Composite - a mixture of two materials to improve properties

- RapiCure resin
- Carbon fiber



Figure 2. Illustration of frontal polymerization propagation.

Frontal polymerization – a process for curing resin, where, after initiation, the heat from the curing reaction is conducted through the material and is sufficient to initiate adjacent monomer. This allows a polymerization front to pass through the part like a wave.

Goal – develop methods to cure a resin system from RapiCure in under 10 minutes with a resulting glass transition temperature of at least 130°C and 90% degree of cure as a composite matrix material.

Impact – composites are growing in popularity for their ability to manufacture parts with very high strength-to-weight ratios and complex curvature. RapiCure has developed commercial frontal polymerization resins, which would lower cost and energy use of composite manufacturing and repair.

Fabrication methods

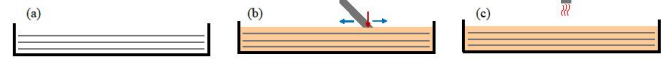


Figure 3. Resin distribution and cure method showing dry fibers in a mint tin (a), mechanical consolidation (b), and heating with a heat gun (c).

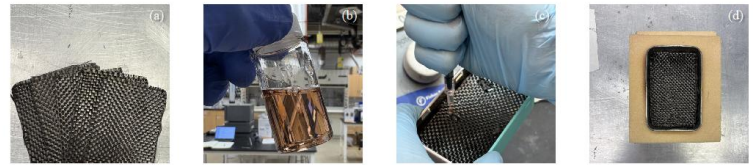


Figure 4. Layout process, showing dry fibers (a), prepared resin (b), fiber saturation (c), and cured composite (d).

Carbon fibers plies were laser-cut and saturated with resin in repurposed mint tins by adding resin one ply at a time. The composites were cured with a heat gun by varying exposure times and distances.

Experiments

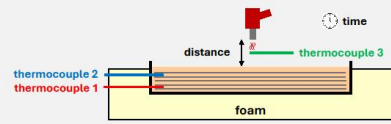


Figure 5. Experimental schematic of thermocouple locations within 5-ply layout.

- 2² designed experiment of time and exposure distance with a midpoint performed on 5-ply composite layouts
- Test optimal conditions for 5-ply composite
- Two layout experiment performed with 15-ply composites

Measurements

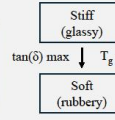
Dynamic mechanical analysis (DMA)

Mechanical properties over a temperature range

- Stiffness (storage modulus)
- Energy dissipation (loss modulus)
- Ratio of loss to storage ($\tan(\delta)$)



Figure 7. DMA fixture.



Differential scanning calorimetry (DSC)

Analysis used to identify thermal events

- Bonds form and release heat during cure
- Residual exotherm of samples compared to complete cure exotherm allows for calculation of degree of cure

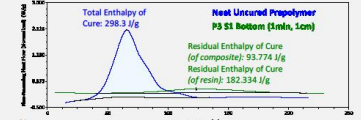


Figure 9. Analysis of DSC thermograms.

Results and discussion

1 5-ply designed experiment

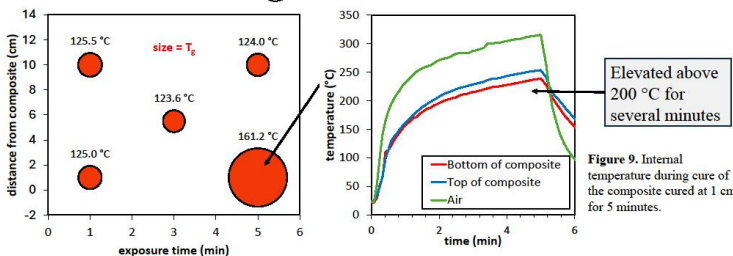


Figure 8. T_g measured by peak in $\tan(\delta)$ plotted against distance from composite and exposure time.

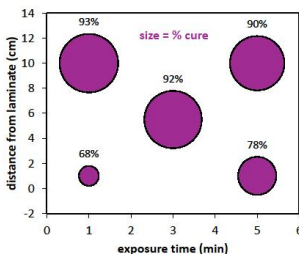


Figure 10. Percent cure measured by peak in $\tan(\delta)$ plotted against distance from composite and exposure time.

Table 1. Linear regression trends between degree of cure (DoC) and factors (exposure time and distance).

	Exposure Time (min)	Exposure Distance (cm)	Interaction Term
T_g (°C)	(+)	(-)	(-)
Cure (%)	None	(+)	None

T_g and percent cure are typically strongly correlated. Why are they not correlated in this experiment?

- The resin mass fraction used to estimate degree of cure could be inaccurate due to resin distribution inhomogeneity.
- The T_g and percent cure are not strongly correlated for this resin at these relatively high conversions.

2 5-ply optimized cure condition

The designed experiment indicated that elevating the internal part temperature above 200 °C dramatically improved the T_g . This could be because the cure leads to a higher T_g when it occurs above the α -relaxation temperature of 155 °C [2].

To achieve a high T_g , the optimal part was cured 1 cm from the heat gun for 3 minutes.

Table 2. DMA from the optimally cured sample. The peak in $\tan(\delta)$ is used to identify the T_g .

Exposure Time (min)	Exposure Distance (cm)	T_g (°C)	Cure (%)
3	1	138.4	98.3

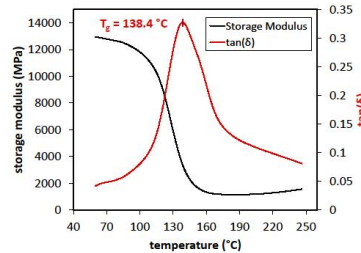


Figure 11. DMA from the optimally cured sample. The peak in $\tan(\delta)$ is used to identify the T_g .

3 15-ply experiments

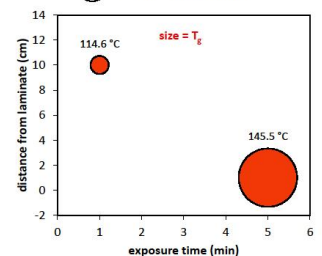


Figure 12. T_g measured by peak in $\tan(\delta)$ plotted against distance from composite and exposure time.

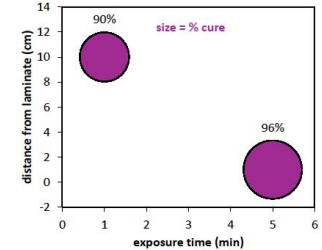


Figure 13. T_g measured by peak in $\tan(\delta)$ plotted against distance from composite and exposure time.

Thick composites were cured at two conditions supporting that a longer exposure time and a shorter distance improve T_g and percent cure.

Conclusion

Composite parts with 5 and 15 plies were manufactured and cured with a heat gun in under 10 minutes. Characterization of these composites revealed that longer exposure times and closer distances improved the glass transition temperature in the composites. From this characterization, we would recommend curing 5-ply composites for 3 minutes, 1 cm away from a heat gun to achieve a glass transition temperature above 130°C and degree of cure above 90%.

Future work

- Analyze the curing conditions for the neat resin.
- Measure the α -relaxation temperature with DSC.
- Measure the T_g with DSC.
- More thoroughly investigate thick samples.

References and Acknowledgments

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